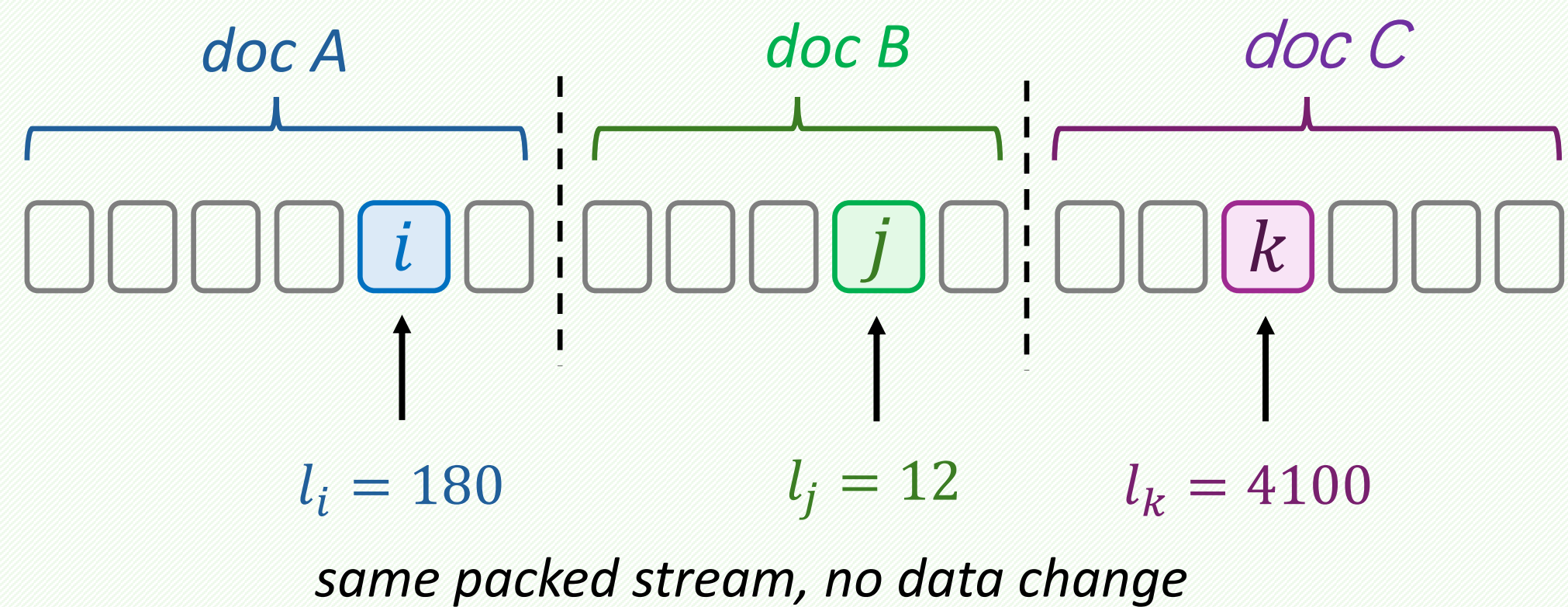
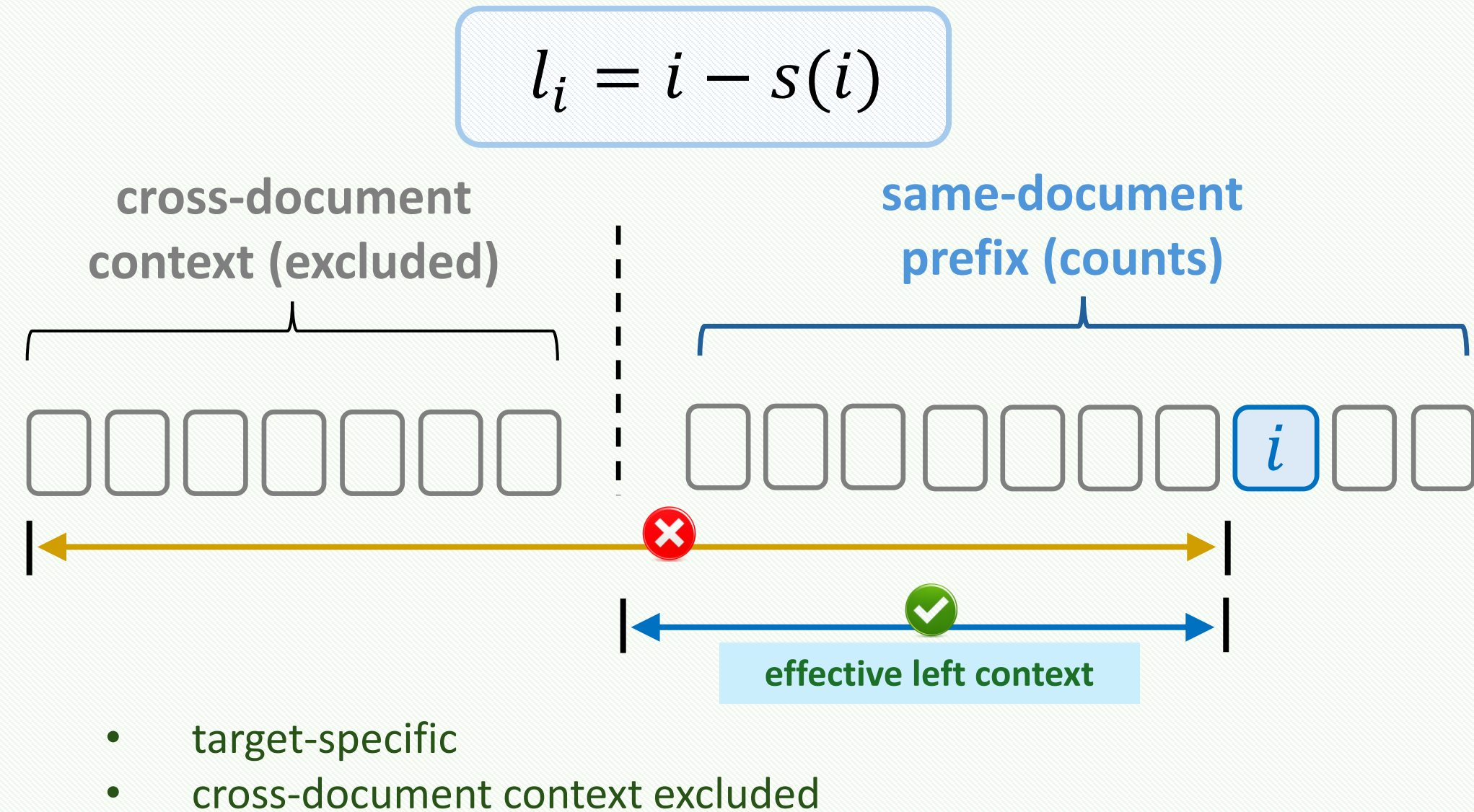


(a) Input and Effective-Context Measurement

1) Packed long-window CPT batch

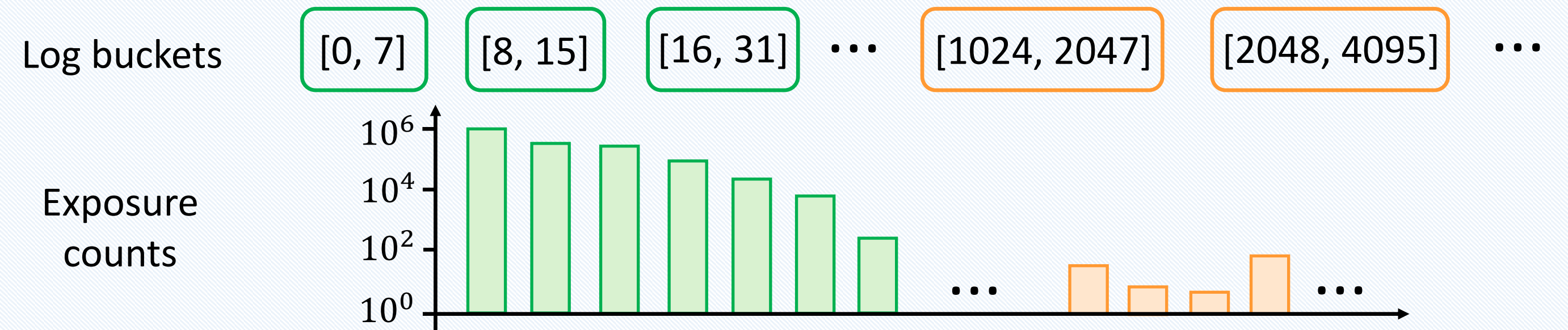


2) Boundary-aware effective left context

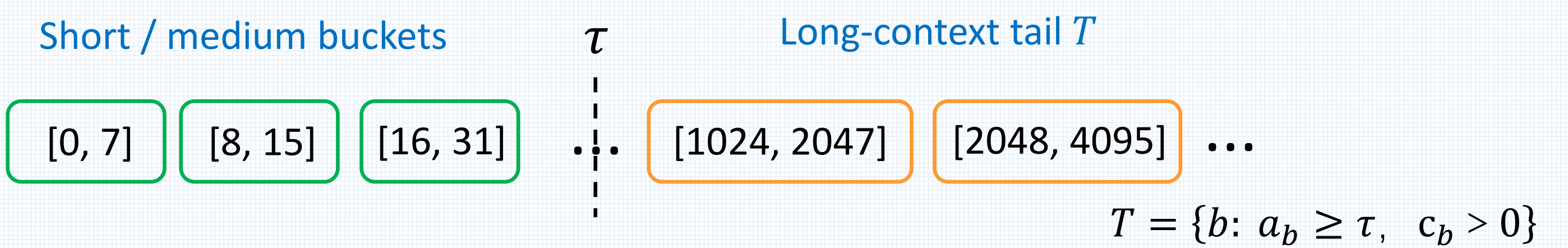


(b) EXACT Supervision Allocation

1) Bucketize target exposure



2) Select long-context tail



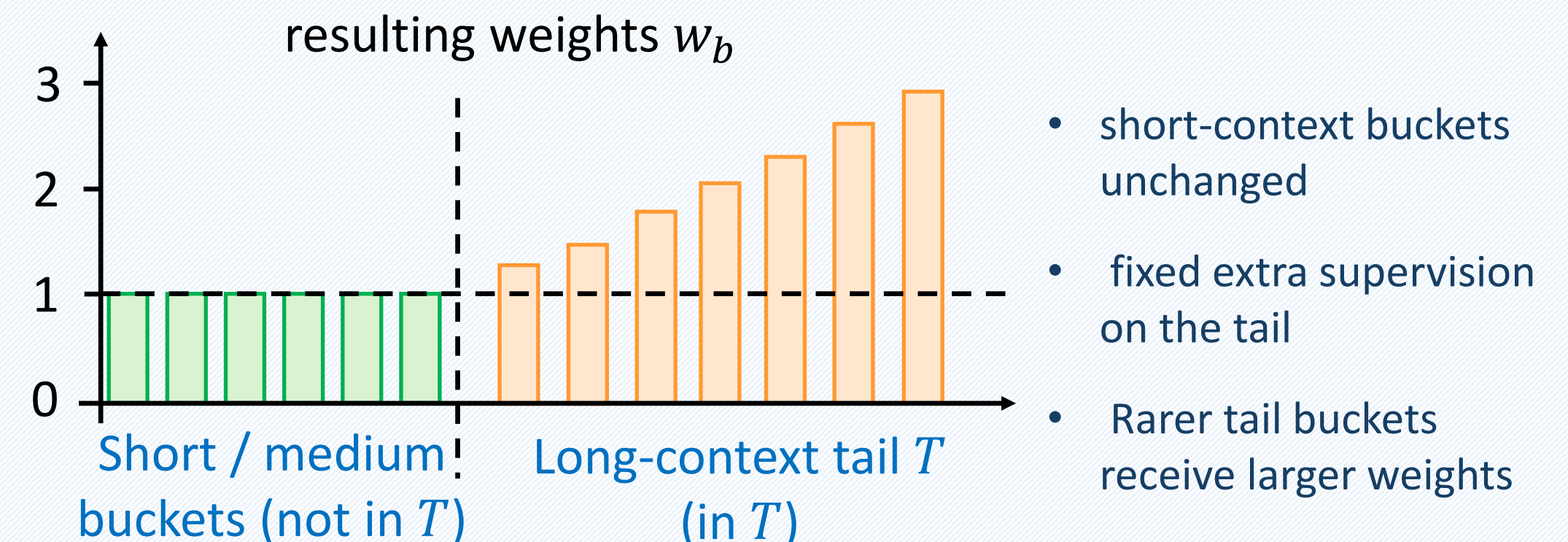
3) Assign Additive inverse-frequency weights

$$q_b = \frac{c_b}{\sum_{i \in T} c_i}$$

$$r_b = (q_b + \epsilon)^{-\gamma}$$

$$w_b = 1, b \notin T$$

$$w_b = 1 + \alpha \frac{r_b}{r}, b \in T$$



(c) Weighted CPT Objective and Output

1) Weighted LM objective

weights $w_{b(i)}$	1	1	1	1.32	1.74	2.61	...
target tokens	x_1	x_2	x_3	x_i	x_j	x_k	...

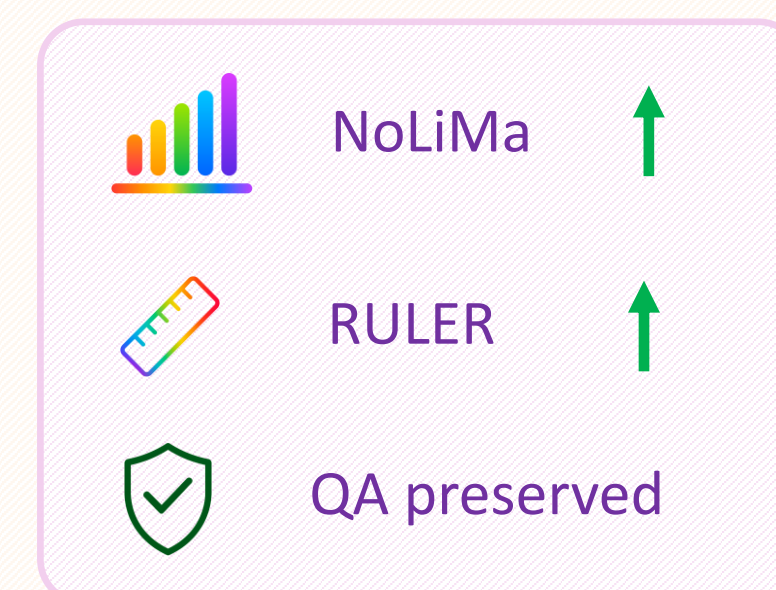
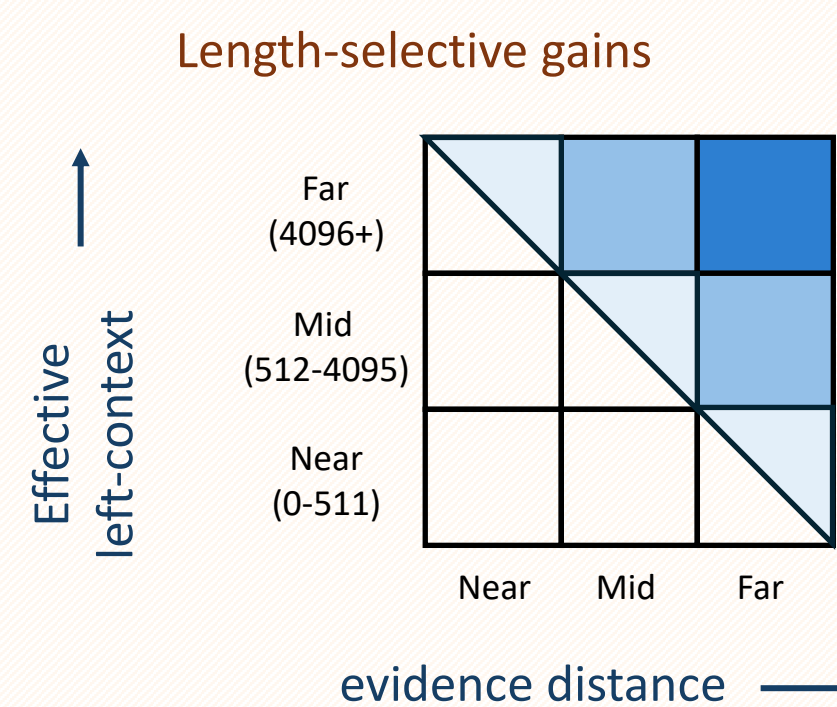
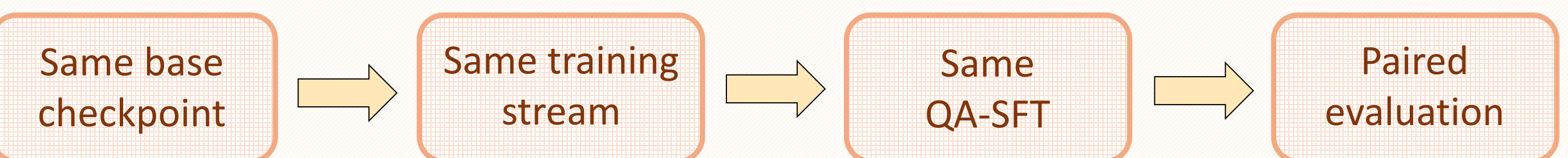
$$\mathcal{L}_{EXACT} = \frac{\sum_i m_i w_{b(i)} CE(x_i)}{\sum_i m_i}$$

m_i : valid same-document mask
 $w_{b(i)}$: bucket weight

2) Continued pretraining

Continued pretraining

3) Paired evaluation and outcomes



- Length-selective gains are strongest in trained-long and extrapolated regimes.